What Does It Cost to Force-Close a Lightning Channel?

Evidence from 974,000 Channel Closes, 2018–2026

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***Abstract*—Bitcoin's Lightning payment network returns the funds of a channel that cannot close cooperatively through pre-signed transactions whose fees and delays fall on its operators. That fee design is being replaced: Bitcoin Core 31.0 removed the mempool rule anchor channels relied on, and the Lightning specification adopted zero-fee commitments in May 2026. What the old design costs has never been measured across fee regimes. We trace every public channel close and every force close on-chain from January 2018 to October 2026, 973,581 closes, and follow each output to its spend. A separate chain index confirms all 698,476 public closes we could compare. Fee spikes did not bring a wave of force closes. Force closes in a spike week and the next were 0.97 times the four weeks before, against 1.02 after ordinary weeks of the same years. A force close costs 2–6 times a cooperative close in fees and locks the closer's balance for days, which outweighs the fees only in low-fee years. When fees are high, pending payments are exposed: in 2023, 18.2% of HTLC timeouts confirmed more than 40 blocks after expiry. Yet since 2021, forwarding nodes lost only 38 payments, each with the preimage already on-chain. Anchor outputs cut late timeouts in fee spikes (20.6% against 30.5%), yet owners rarely bump commitments with them, and miners take most of what third parties sweep. Zero-fee commitments would save only 12% of the anchor mechanism's block space; they should be judged by how promptly payments resolve when fees spike.**

***Index Terms*—Lightning Network, Bitcoin, payment channel networks, force close, transaction fees, anchor outputs, fee bumping, zero-fee commitments, HTLC, measurement**

I. Introduction

The Lightning Network is a payment network built on Bitcoin. Its nodes hold their operators' funds in two-party payment channels, each backed by one output on the Bitcoin chain. When both parties agree, a channel closes cooperatively with one ordinary transaction. When they cannot, because a peer is offline, a payment is stuck or the two implementations disagree, either party can force-close the channel: it broadcasts the latest commitment transaction, signed in advance by both, and then claims its share through further pre-signed and time-locked transactions [1]–[3]. The fees and delays of those transactions fall on the operators, and a transaction that confirms too late can lose a pending payment or let a cheating peer win [4]–[6]. The fee design of these transactions is changing now. Bitcoin Core 31.0 (April 2026) removed the CPFP carve-out, the mempool rule that anchor channels were designed around [7], and in May 2026 the Lightning specification adopted zero-fee commitments, which rely on newer relay rules instead [8]–[10]. This paper measures the old design, across the fee regimes it lived through, before the migration, as a baseline for judging the new one.

Force closes have been counted [11]–[13] and predicted [14], but what they cost has not been measured across fee regimes. The questions that shaped the design are argued in developer forums from a few nodes' logs or from none: whether owners use their anchor outputs to bump fees, whether third parties sweep them at a profit, and whether fee spikes force channels closed [15], [16]. Our previous paper measured the cost of force closes over a few months of low fees in 2026 [17].

We measure it from the chain. We follow every close of every public channel since Lightning's first channels in January 2018, find the force closes of channels outside the public graph by the fingerprint of their commitment transaction, and follow each of their 1,346,588 outputs to the transaction that spent it: 973,581 closes in all, 523,815 of them force closes. Fee statistics for every block, from our own node, place each transaction in the fee market of its block. The main answer is that fee spikes do not bring a wave of force closes. In the week of a spike and the week after, force closes were 0.97 times their level in the four weeks before, while ordinary weeks of the same years show 1.02. That holds for pre-anchor channels, whose commitment fee is fixed when it is signed, as for anchor channels, whose fee can be raised at broadcast. The cost of a force close lies elsewhere: in its ratio to a cooperative close, in the days the closer's funds stay locked, and in the exposure of pending payments when fees are high. The wait outweighs the fees only in low-fee years, and the exposure has rarely become a loss. We make five contributions.

- **A census with rates.** Force closes became rarer per channel, and private channels now produce most of them. Of the public channels funded in 2019, 35% were force-closed within a year; of those funded in 2025, 17%. The force share of public closes fell from 44% in 2023 to 20% in 2026. In 2026 force closes of channels outside the public graph outnumbered public ones 3.4 to one (§V).
- **The cost of a force close in fees, time and block space.** A force close costs 2–6 times a cooperative close in fees and, since 2022, about three times its block space; the closer waits a median of 1.7 days for its own balance, which outweighs the fees only in the low-fee years 2025–

26. Counting the open as well, a public channel's whole life cost a median of 3,435 sat on-chain. When fees are high, HTLC timeouts confirm late, and anchor channels' timeouts less often than pre-anchor ones. Yet forwarders lost few payments: 38 since 2021, all with the preimage on-chain first. The only losses in the order a replacement-cycling attack leaves were a test that one operator ran against its own node, and 2026's one wave of penalties matches a splicing bug already fixed in Core Lightning (§VI).

- **The anchor money trail.** Anchor outputs became third-party dust. Owners spent 81% of anchors in 2022 and 5% in 2026; third parties sweep the rest, from 2025 at the first block they can, and fees took 87% of the swept value in 2026. No P2WSH anchor was left unspent, while no third party has swept a simple-taproot anchor, so the premise of an open LND proposal to stop sweeping anchors holds only for the older kind (§VII).
- **Fee regimes.** Fee spikes did not bring a wave of force closes, judged against placebo weeks of the same years. Fixed commitment fees overpaid in calm markets, and all of Lightning's closes and required spends took at most 0.18% of Bitcoin's transaction fees in any year (§VIII).
- **The first zero-fee commitments.** The first zero-fee commitment closed on 29 October 2025, before the specification was merged. Across the anchor force closes we observe, the new format would have saved 12% of the anchor mechanism's block space (§IX).

This is the third paper of a programme on the Lightning Network's resilience. The first asked whether client diversity protects the network when one implementation fails [17], the second who patches it [18], and this one what a failed channel costs on-chain. All data behind the figures, tables and in-text numbers are released with the code. We report aggregates only: no node, address or transaction is named.

II. Background and Related Work

A. Force closes

A channel is a 2-of-2 multisignature output on the chain. Each party holds its own version of the latest commitment transaction, which spends that output to the two balances and to one output per pending payment (HTLC). In the holder's version, the holder's own balance (`to_local`) is locked for `to_self_delay` blocks, a value the peer chose at opening, while the peer's balance (`to_remote`) is not [2]. The delay gives the peer time to react if the holder broadcasts an old, revoked commitment: the peer can then take every output with a penalty (justice) transaction [1]. A force close is the broadcast of a commitment; the broadcaster waits out the delay before sweeping its balance.

Pending payments add deadlines. An offered HTLC returns to the sender after its expiry height (`cltv_expiry`) through a pre-signed HTLC-timeout transaction, or, on the peer's commitment, directly; a received HTLC is claimed with the payment's preimage through an HTLC-success transaction [2], [3]. A node that forwards a payment holds two HTLCs for it, and the incoming one expires a margin later than the outgoing one, the `cltv_expiry_delta` the node advertises in its channel updates [19]. If the node cannot time out its outgoing HTLC on-chain within that margin, the downstream node can claim it with the preimage while the upstream one times out the incoming HTLC, and the forwarder pays for a payment it is not paid for. Flood-and-loot, time-dilation and replacement-cycling attacks all work by making such a transaction confirm too late [4]–[6].

B. Who pays the fee

Commitment transactions are signed when the channel state changes and broadcast later, at a fee rate nobody knew at signing. Three formats answer this differently.

Pre-anchor commitments carry a fee fixed at signing. The channel's opener proposes it with `update_fee`, and a peer that finds the rate unreasonable can fail the channel [20]. The broadcaster cannot raise the fee later, because its own output is time-locked. These disagreements are a commonly cited cause of force closes when fees move fast.

Anchor commitments (specified in 2020 [21]) add two 330-sat anchor outputs, one per party, which their owner can spend at once with a child transaction that pays for both (child pays for parent, CPFP). Sixteen blocks after the commitment confirms, anyone can spend either anchor, so that unused anchors do not stay in the UTXO set [2]. The peer's `to_remote` output carries a one-block delay, so that the anchors are the only outputs a child can spend at once, and the CPFP carve-out, a Bitcoin Core mempool rule, let each party attach one child even if the other had filled the package limits [7]. Second-stage HTLC transactions are signed so that the holder can add inputs to pay their fee [2]. Simple taproot channels follow the same design with taproot outputs and anchor scripts of their own [22].

Zero-fee commitments [8] pay no fee at all. The commitment is a version-3 (TRUC) transaction [9] with one keyless pay-to-anchor (P2A) output in place of the two anchors, and it is relayed together with a child that pays for both, which Bitcoin Core supports since version 28.0 [10]. TRUC's topology limits replace the carve-out as the defence against transaction pinning [23], and Bitcoin Core 31.0 removed the carve-out [7]. The specification merged the format in May 2026; channels move to it as operators upgrade their software, which in the Lightning Network takes months to years [18].

C. Related work

BitMEX Research first counted non-cooperative closes from the chain, private channels included, and found more than 60,000 by January 2020 [11]. Zabka et al. classified how public channels close [12]; Grötschla et al. followed public channels from funding to the spending of their closing outputs [13]; Antonelli et al. predict which channels will close and how [14]. Kappos et al. and Romiti et al. used closing transactions to link channels to on-chain wallets [24], [25]. Our previous paper measured force closes around the 2026 Core Lightning

shutdown [17]. None of these accounts for what a force close costs or follows the anchor outputs.

The fee-bumping design has been argued from small samples. Daftuar analysed the 2023 transactions that spent anchor-like outputs, to test whether Lightning's commitment spends would fit version-3 relay rules [15]. An open LND issue proposes to stop sweeping "dangling" anchors, on the argument that third parties sweep them more efficiently, and gives the break-even fee rate without data [16]. The security literature models what happens when closing transactions confirm late: mass closes that congest the chain [4], [26], nodes kept from seeing new blocks [5], pinned or cycled replacements [6], [23], HTLC slots jammed by attackers [27] and fees taken from revoked commitments [28]. We measure how often, and in what fee conditions, such deadlines are missed in practice.

III. Data

a) The chain: We scan every block from block 503,816, before Lightning's first channels, to block 974,041, the last block mined before 1 November 2026 00:00 UTC, with our own Bitcoin Core node. For every block we keep its fee statistics (`getblockstats`: the size-weighted percentiles of fee rates, total fees and total weight). A full index of the chain built from the same node, with every transaction's locktime and every input's witness, gives the HTLC expiries and the funding fees used in §VI-C and §VI-D. On the blocks from 2021, which we also queried from the node directly, its fee totals agree with `getblockstats` on every block.

b) Public channels: Our list of public channels joins our own gossip since May 2023, mempool.space's record of closed channels and their close reasons [29], and the lnresearch gossip archive for the earlier years [30]. A short channel ID encodes the block, transaction and output of the channel's funding output, so the list resolves to 736,110 funding outpoints, each to the transaction that spends it: the close. The list holds few of the channels that closed in 2018: the gossip archive and mempool.space's record become dense only in January 2019, and only about 1% of 2018's force closes are of listed channels. For 2018 we therefore report force closes only, listed or not, which the commitment fingerprint finds completely; public, cooperative, force-share and per-channel series start in 2019, and January 2019 is still incomplete.

c) Closes and their outputs: A close is cooperative or forced; a close that funds a new channel in the same transaction is a splice and is left out. Force closes of channels that are not in the list are found by the fingerprint of the commitment transaction alone (§IV); we call them *unlisted*. Up to the cutoff the data hold 699,049 closes of public channels (449,707 cooperative, 249,283 forced) and 274,532 unlisted force closes. For each of the 1,346,588 outputs of these 523,815 force closes we record the transaction that spent it, if one had by the cutoff (1,212,596 did), and classify the spend from its witness script: the broadcaster's delayed sweep, the peer's sweep, an HTLC transaction or claim, an anchor spend by its owner or by a third party, or a penalty. Statistics about what happens to a close's outputs use only closes confirmed at least 4,032 blocks (four weeks) before the cutoff, so that their outputs had time to be spent.

d) Validation: We check the closes two ways. First, the chain index above is a separate extraction of the chain, by a different program: in it, the transaction that spends each public channel's funding output is the close we recorded, for all 698,476 closes that could be compared, and its shape, with or without the commitment fingerprint, matches our force or cooperative label in every case. Second, mempool.space gives a close reason for 69–94% of the public channels that closed in each year from 2019 to 2025, and none for 2026. It labels closes from the chain as well, so agreement with it is a check of consistency, not independent validation. Where it gives a reason, for 480,440 of our public closes (69%), our label agrees on 100.00%.

e) Client labels: For the appendix we attach to each public close the implementations at its two ends, using the labels and classifier of our first paper on the day of the close [17]. They exist from May 2023 and inherit that paper's accuracy.

f) What the chain does not show: The chain does not show which party broadcast a force close, so every cost is per channel, both parties together. It shows fee bumps that confirmed, not those that were tried and failed. Private channels appear only when they force close: their cooperative closes look like ordinary payments, so their number and their force-close rate are unknown. §XI lists the consequences.

IV. Methods

A. Force closes and their formats

A commitment transaction carries a fingerprint: the upper byte of its locktime is `0x20` and that of its single input's sequence is `0x80`, because the two fields encode the obscured commitment number [2]. We call a transaction a force close when it has this fingerprint and spends a 2-of-2 multisignature output or a taproot output by its key path. Its format is *zero-fee* if it is a version-3 transaction with a pay-to-anchor output, *simple taproot* if it spends a taproot output, *anchor* if it has at least one anchor output, and *pre-anchor* otherwise.

B. Rates and cohorts

Counts of closes follow the size of the network, so we also report rates. A public channel is *open* at a height when its funding output is still unspent. Channels whose nodes have left the network count as open until they are closed on-chain, which is right for a rate of on-chain closes but inflates the denominator as such channels accumulate. The force-close rate is the number of public force closes per 1,000 open public channels per month (or week). Following our first paper [17], a *mass-close event* is a node party to at least 20 public force closes on one UTC day; *organic* force closes are the others. The *cohort incidence* at age a is the share of the public channels funded in a year that were force-closed by age a, among those funded at least a before the cutoff. Open channels do not enter it, so it is not biased by abandoned channels; we draw a curve while at least half of its cohort is observed.

C. The cost of a force close

We count every transaction a force close requires, in three currencies. *Fees*: the commitment's fee; an anchor spend by its owner in the commitment's block, a CPFP fee bump, counted in full; HTLC transactions and claims; the broadcaster's delayed `to_local` sweep; the `to_remote` sweep of anchor channels, which the one-block delay makes a separate step; and later anchor spends by their owner, which we call cleanup. A spend that batches several inputs is split equally among them; charging each close the whole transaction gives an upper bound. Not counted: the `to_remote` spend of a pre-anchor channel, a plain wallet output like those of a cooperative close; third-party anchor sweeps, which are paid from the anchors' own 330 sat; and penalties, reported apart. Each year's median is set against the median fee of a cooperative close in the same year. *Time*: the days from the close until the broadcaster's `to_local` can be swept, from the `to_self_delay` revealed in its witness, and the balance times that wait (BTC-days). To set the wait against the fees without assuming a cost of capital, the break-even rate is the yearly return on the locked balances at which a year's BTC-days would cost as much as that year's force-close fees. *Block space*: the virtual bytes of the same transactions.

D. HTLC deadlines

The chain gives each HTLC's expiry: an HTLC-timeout transaction's locktime is the expiry, because its pre-signed signature commits to it, and a received HTLC's script, revealed when it is spent, contains it [2]. We measure how many blocks after its expiry each HTLC-timeout transaction confirmed and, for HTLC-success transactions, how many blocks before. The yardstick is the margin a forwarding node keeps, its `cltv_expiry_delta`: we use 40 blocks, the tenth percentile of the values public channels advertised in our daily gossip snapshots from May 2023 to the cutoff, and 80, their median. The fee condition is the median fee rate of the block at the expiry. To compare formats we use the HTLCs whose commitment had confirmed by the expiry, so that the delay is the HTLC transaction's own. Both HTLC scripts also contain a hash of the payment hash; until point time-locked contracts replace hashes, every hop of a payment and every part of a multi-part payment share it, so the same hash in two force closes marks one payment.

E. Lifecycle cost

For each public channel closed at least four weeks before the cutoff, the lifecycle cost adds the channel's share of its funding transaction's fee to the cost of its close and required spends (as above, or the fee of the cooperative close). A funding transaction's fee is split equally among the channels it opens; its other outputs count as the opener's change and pay nothing, because the change exists only because of the open. A splice is charged once, to the new channel's open. Rules that also charge change outputs, by value or by size, are reported as a robustness check.

F. Anchors and who sweeps them

Each P2WSH anchor is spent by its owner in the commitment's block (a bump), by its owner later (cleanup), by a third party (an input with an empty signature, possible from the sixteenth block), or not at all. For a same-block owner spend we also test whether the child raised the fee rate of the package, which separates a bump from cleanup that happened to confirm in the same block. Third-party sweeps are collected chain-wide, including anchors of commitments we never saw, and each sweep's 330 sat per anchor is split between the fee (to miners) and the outputs (to the sweeper). The break-even fee rate, at which the fee for one anchor input equals the anchor's value, is 6.38 sat/vB. To measure concentration we link sweep transactions that spend each other's outputs or pay the same address; each connected group is an operator. Sweeps whose outputs are all `OP_RETURN` give everything to miners and are counted apart. We report shares only and name no operator or address.

G. Fee spikes

To test whether fee spikes bring force closes we use an event study and a regression. A *spike week* has a median block fee rate at least 2 times that of the previous eight weeks and at least 10 sat/vB; only the first week of a cluster counts. We compare organic public force closes in the spike week and the week after with the four weeks before, pooled over all spikes. The regression, by week, relates the logarithm of organic public force closes per open channel to the logarithm of the median block fee rate, with year fixed effects and a linear trend, so that weeks are compared within the same fee regime rather than 2023 with 2025. We fit it with the fee rate of the same week, of one to four weeks earlier and of one week later.

We judge both against placebos that keep the force-close series as observed and break only its timing against fees. (A) For each spike, an ordinary week of the same year is drawn at random, away from every real spike (10,000 draws). (B) Each year's fee series is cut into 8-week blocks, which are shuffled within the year (5,000 draws). (C) The fee series' deviations from its year means and trend are rotated by every shift of at least 13 weeks (382 shifts), which keeps all of its autocorrelation. A result's percentile is the share of draws below it. The random seed (42) and the verdict rules were fixed before the full run: a wave counts only if the spike ratio is above 95% of placebo A, and a delayed rise only if the largest coefficient of lags one to four beats the largest in each draw at $p < 0.05$ under both B and C. We do not use Newey–West intervals: under the placebos, nominal 5% tests rejected in 12–14% of draws of B and 21–25% of draws of C.

Fee guessing. A commitment's fee rate is compared with the median and the tenth percentile of the fee rates in its own block. *Sweep timing.* A third-party sweep's block is compared with the block sixteen blocks after the close, when the sweep first became possible. *Closing bill.* The fees and block space of every close and required spend, public or not, plus third-party sweeps, are compared with all transaction fees and block weight in each year.

H. Zero-fee counterfactual

For every anchor force close we compare the block space of the anchor mechanism, two 43-vB anchor outputs and the transactions that spend them, with that of a zero-fee commitment: one 13-vB pay-to-anchor output, spent in the same package, and a child with a wallet input and change for the closes that were not already bumped. An owner's anchor input takes 69.75 vB, a third party's 51.75 vB (measured on mainnet sweeps) and a pay-to-anchor input 41.25 vB.

V. How Often Channels Force-Close

a) Force closes became rarer, but less than their counts suggest: Public force closes peaked at 57,896 in 2022 and fell to 15,440 in the first ten months of 2026 (Table I). Part of that fall is a smaller network: open public channels peaked at 90,340 and stood at 40,353 at the cutoff. Part is more cooperative closing: cooperative closes released 8,885 BTC of capacity in 2024 and 19,796 BTC in 2025. The force share of public closes therefore fell faster than the risk to a channel. It went from 44% in 2023 to 20% in 2026, while the organic force-close rate went from 41 to 26 per 1,000 open channels a month (Fig. 1c).

b) 2019 reads highest, but our list may be incomplete: In 2019 the force share (50%) and the organic rate (50 per 1,000 open channels a month) were the highest of any year. We do not read this as a peak, because our channel list may still have been filling in. In February and March 2019 the organic rate was 106 and 103 per 1,000, and no later month exceeded 64.

c) The risk to a new channel fell: 35% of the public channels funded in 2019 were force-closed within a year, against 17% of those funded in 2025 (Fig. 2). The fall was not steady: the share dropped to 24% for the 2021 cohort, rose to 31% for 2023's and fell again after. Cohort incidence does not depend on the count of open channels, which abandoned channels inflate, so it is the cleaner measure of the trend.

d) Much force closing is one node closing many channels at once: Every year has 64–239 mass-close events, which hold 19–32% of public force closes; the largest put 856 channels on-chain in one day. Without them the rate still falls, so the decline is not one operator's. The 2026 peak is such an episode. In August 2026 the rate reached 51 per 1,000 open channels, 31 without mass-close events, while open channels fell from 50,219 in July to 40,353 at the cutoff. That month brought the Core Lightning shutdown and the largest contraction of the public graph. Our first paper found the contraction driven by large operators leaving, and organic force closes of channels with a Core Lightning node above their usual rate after the shutdown [17].

e) Most force closes now happen outside the public network: In 2026, unlisted force closes outnumbered public ones 3.4 to one. From 2024 on they are almost all private channels; before 2024 they include public channels missing from our list (§XI). Private channels are often those of mobile wallets [11], and 23% of 2026's unlisted force closes were simple taproot channels. The largest burst in the data is unlisted: on 1 September 2024, 8,990 unlisted force closes holding 36.7 BTC confirmed within about 2 hours (18 blocks), 6,910 of them pre-anchor.

f) Most of 2026's pre-anchor closes are one cluster: Pre-anchor commitments were 87% of public force closes in 2021 and 26% in 2026 (Table I), but most of the 2026 share is one cluster. Five nodes were party to 74% of 2026's pre-anchor public force closes, four of them LDK nodes that announced no anchor support [17], and their channels lasted a median of 4 days. Without closes involving those nodes, pre-anchor commitments were 8% of 2026 public force closes, and those channels were old (median 3.9 years).

g) Client pairs: Among client pairs with more than 1,000 closes since May 2023, the force share ranges from 28% (LND–LND) to 40% (CLN–CLN), with one outlier that a single node explains (Appendix A; labels from [17]).

VI. What a Force Close Costs

A. Fees, time and block space

a) Cheap in fees, a multiple of a cooperative close: In fees alone a force close is cheap: the median public force close cost 925 sat in 2026 (Table II). The median was highest in 2020, at 14,318 sat, against 7,346 sat in 2023. Against a cooperative close of the same year it is dear: 2–6 times the median cooperative close in every year. The fee market sets the level, not the ratio. Force closes paid 31.1 BTC in fees over the whole period, 11.97 BTC of it in 2023, more than in any other year.

b) When fees are high, the money goes to bumps and HTLCs: In 2023, fee bumps took 34% of all force-close fees and HTLC resolution 14% (Fig. 3b). That year 44% of anchor force closes were bumped by a child in the commitment's block, and 78% of those with pending HTLCs. In the low-fee years the commitment and the sweeps make up most of the cost.

c) Anchors doubled the block space: An anchor force close uses about twice the block space of a pre-anchor one (median 639 against 292 vB): two extra outputs, a `to_remote` output that needs a sweep of its own, and the anchors' spends. Since 2022 a force close has taken 3.0–3.1 times the block space of a cooperative close. Block space does not depend on the fee regime, which makes it the currency in which formats compare most directly (§IX).

d) The closer waits days: The broadcaster's own balance returns a median 1.7 days after the close, and in the slowest tenth of closes up to 22 days (Fig. 3c). The wait is the `to_self_delay` the peer chose at opening: 49% of values are 144 blocks (one day) and 10% about two weeks. Over the period, closers' balances were locked for 68,036 BTC-days.

e) The wait outweighs the fees only when fees are low: The break-even rate was 2.7% in 2025 and 1.2% in 2026, so in the low-fee years even a modest return on the locked funds made the wait the larger cost. Before 2025 it was 6.1–54.1% a year. Over the whole period the wait cost more than the fees only for funds earning more than 16.7% a year.

TABLE I

LIGHTNING CHANNEL CLOSES ON THE CHAIN, 2018–2026. ALL FORCE: EVERY FORCE CLOSE, PUBLIC OR NOT. PUBLIC: CHANNELS IN OUR LIST OF EVERY PUBLIC CHANNEL. UNLISTED: FORCE CLOSES OF CHANNELS OUTSIDE IT, FOUND BY THE COMMITMENT PATTERN.

Year	All force	Coop.	Force	Force %	Force BTC	Age (d)	Unlisted	Unl. BTC	Pub. legacy %	Unl. taproot %
2018[b]	35,218	–	–	–	–	–	–	–	–	–
2019[c]	52,148	30,404	30,001	49.7	679	80	22,147	411	100.0	0.0
2020	29,877	28,785	17,813	38.2	581	159	12,064	198	100.0	0.0
2021	39,539	64,048	27,393	30.0	1,123	94	12,146	179	87.5	0.0
2022	73,569	92,645	57,896	38.5	3,226	133	15,673	264	44.8	0.0
2023	76,869	62,481	49,998	44.5	3,363	151	26,871	764	40.2	0.1
2024	76,634	50,874	28,888	36.2	2,208	215	47,746	1,410	33.9	0.6
2025	71,498	58,137	21,518	27.0	1,975	242	49,980	838	23.4	3.2
2026[a]	68,463	62,311	15,440	19.9	2,279	142	53,023	974	25.6	23.1

[a]January–October. [b]From 12 January (block 503,816). Our list of public channels holds few of the channels that closed in 2018, so only the force-close total is shown. [c]The list's coverage is still ramping up in January 2019. Age: median age at force close. Pub. legacy: share of public force closes with a pre-anchor commitment. Unl. taproot: share of unlisted force closes of simple taproot channels. Splices (59) are not shown.

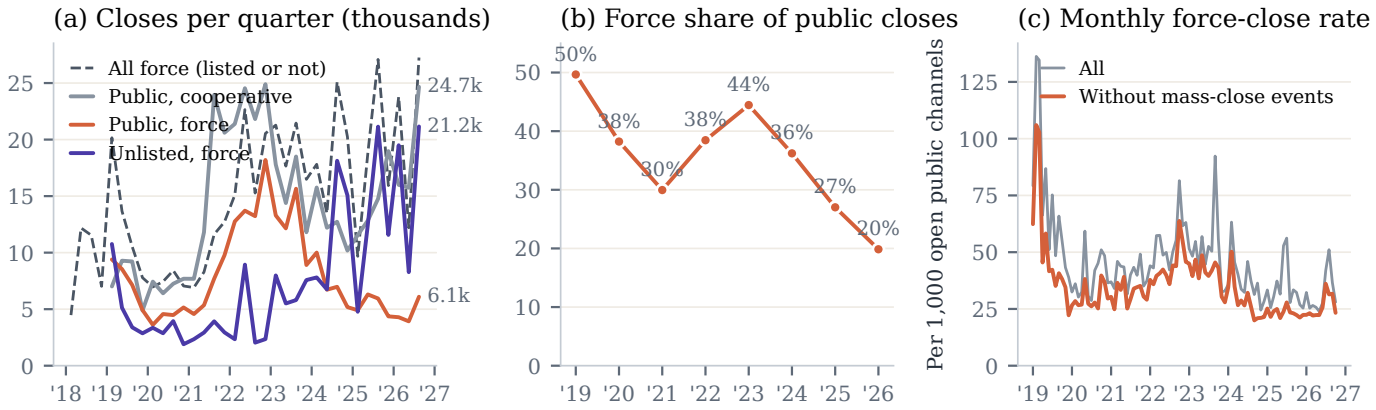


Fig. 1. Lightning channel closes on the chain. (a) Closes per quarter: all force closes, listed or not (dashed, from January 2018), and from 2019 the cooperative and force closes of public channels and the force closes of channels outside our list of public channels (unlisted). (b) Force share of public closes by year, from 2019. (c) Public force closes per 1,000 open public channels per month, from 2019, with and without mass-close events (one node party to at least 20 force closes on one UTC day).

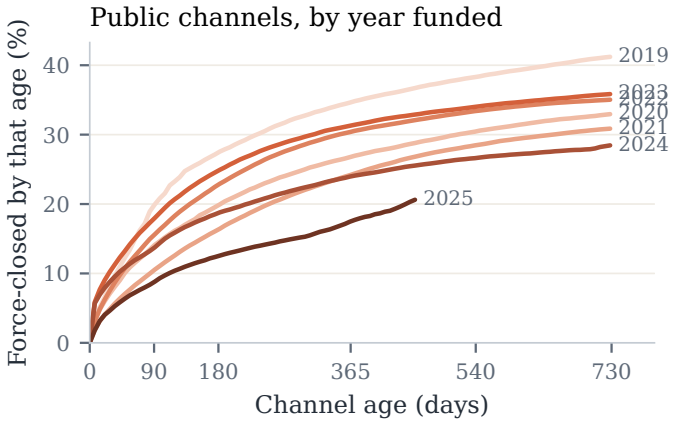


Fig. 2. Share of public channels force-closed by each channel age, by the year the channel was funded (2019–2025). A curve ends where less than half of its cohort has been observed for that long.

f) Relative to capacity, the cost fell: In 2021, 41% of public force closes paid more than 1% of the channel's capacity in fees; in 2026, 8%.

B. Penalties and a breach wave

a) Penalties are rare: Over the period, 950 penalty transactions took 11.4 BTC from parties that broadcast a revoked commitment, in public or private channels. Until September 2026 a typical month had 2 and none had more than 61 (September 2018). Only punished breaches are visible: a breach that went unpunished looks like an ordinary force close.

b) A one-week breach wave: Between 28 September 2026 and 5 October 2026, revoked commitments of 306 private anchor channels were broadcast, and every one was penalized within 120 blocks. 305 of them (3.18 BTC) were splices of channels opened a median of 5 blocks earlier by one funder, which also collected the penalties. The counterparty force-closed about 3 blocks after each splice; 90 more such closes that week used the current state. The remaining channel is an unrelated breach of an older channel. The transactions on both sides follow Core Lightning's construction rules, and the wave matches a bug that Core Lightning fixed in version 26.06.8, released on 19 September 2026 [31]. In affected versions, a unilateral close after a splice had locked broadcast the commitment from the moment of the lock, which is revoked as soon as the channel is updated again. Core

TABLE II
WHAT A PUBLIC FORCE CLOSE COSTS, AGAINST A COOPERATIVE CLOSE, 2019–2026 (MEDIANS, SAT UNLESS STATED).

Year	Force closes	Commit.	Total	p90	Coop.	Ratio	vB	Coop. vB	¿1% cap.	Days
2019	30,001	4,900	7,460	24,802	3,485	2.1	270	138	54.9	1.0
2020	17,813	9,051	14,318	36,545	3,994	3.6	271	168	63.1	1.0
2021	27,393	2,583	3,984	31,397	697	5.7	302	168	41.0	1.0
2022	57,896	2,021	3,342	7,920	551	6.1	521	169	16.1	1.7
2023	49,998	2,872	7,346	45,588	2,005	3.7	595	192	32.1	2.1
2024	28,888	3,012	6,332	36,635	2,124	3.0	581	192	29.6	2.0
2025	21,518	730	1,292	3,912	262	4.9	598	192	11.7	1.7
2026[a]	14,435	597	925	2,363	197	4.7	585	192	7.5	2.5

[a]Closes to 4 October 2026, so that outputs had four weeks to be spent. Total: every fee the force close requires (commitment, fee bump, HTLC resolution, sweeps, anchor cleanup); batched spends split per input. Ratio: median total over the median cooperative close fee. vB: block space, the same way. >1% cap.: share of force closes whose fees exceed 1% of channel capacity. Days: median wait before the broadcaster's own balance can be swept.

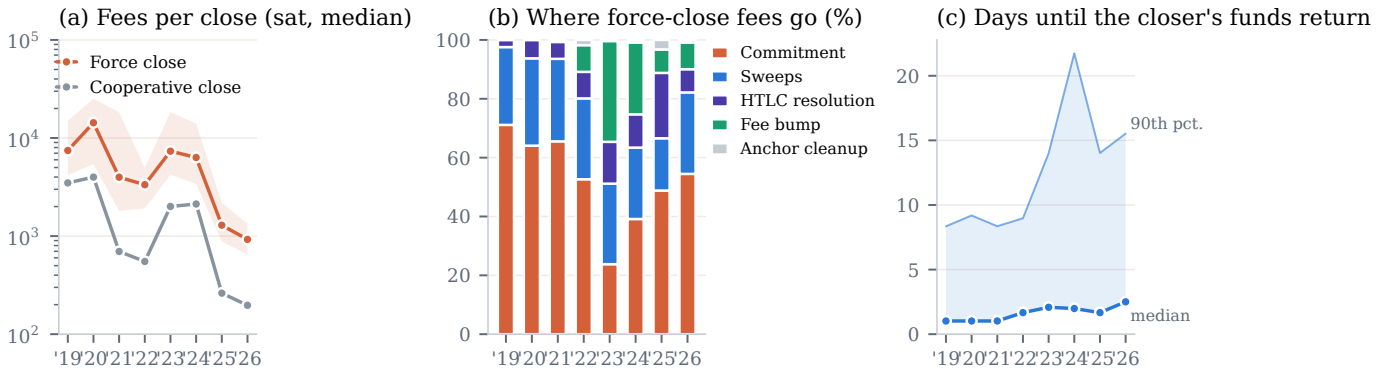


Fig. 3. The cost of a public force close. (a) Median fees per close, force and cooperative (log scale; band: interquartile range of force closes). (b) Where force-close fees go: the commitment, sweeps of the broadcaster's balance and of anchor channels' `to_remote`, HTLC resolution, fee bumps (owner CPFP children in the commitment's block) and later anchor cleanup. (c) Days until the broadcaster's own balance can be swept: median and 90th percentile.

Lightning locks a splice at three confirmations, which is when the closes came. The bug explains why the old states were always post-splice states and why some closes used the current state. The chain does not show what made each counterparty close at the lock, or whether the funder caused it. This one week holds most of 2026's 321 penalties and is the only event of its kind in our data, which go back to 2018.

C. Exposure: HTLC deadlines

a) Most HTLC timeouts are prompt; the late tail grows with fees: Of 70,808 HTLC-timeout transactions, the median confirmed 1–3 blocks after its expiry in every year. The tail is what matters for a forwarding node. More than 40 blocks late were 6.5–8.0% of timeouts in each year from 2019 to 2022, then 18.2% in 2023, the high-fee year, and 16.4% in 2024. In 56% of cases the commitment had confirmed before the expiry, so the delay was the HTLC transaction's own.

b) Anchors keep HTLC timeouts closer to the deadline when fees are high: When the block median fee rate at expiry was 50 sat/vB or more, 30.5% of pre-anchor HTLC timeouts confirmed more than 40 blocks late, against 20.6% of anchor ones (Fig. 4a). At 5–20 sat/vB the gap was 13.2% against 4.7%; below 5 sat/vB the two formats were alike. A pre-anchor HTLC transaction carries the fee fixed at signing, while an anchor channel's can add inputs to pay a fee chosen at broadcast. This is the effect the anchor design was built for, and unlike the commitment-level test of §VIII it shows up in the data. The exception is 2024, when anchor channels' timeouts were late more often than pre-anchor ones (12.8% against 6.3%).

c) Successes mostly come early: HTLC-success transactions confirmed a median 8–19 blocks before expiry in every year. In most years only 1.0–2.4% came at or after the expiry, when the peer can already take the HTLC back. Three years had more: 16.0% in 2024, 8.6% in 2019 and 8.5% in 2023.

d) One payment can take several channels on-chain: The payment hash links one payment's HTLCs across closes (§IV-D). Of the force closes with an HTLC resolved on-chain, 25% share a payment with another force close (Fig. 4b): 6,951 payments, up to 18 closes for one payment, with a median of 140 blocks between the first and the last. 96% of them were timed out on every channel: a stuck payment, or the parts of one multi-part payment, taking several channels down along its route.

e) Split resolutions are rare losses: From 2021, in 40 payments a forwarding node's two HTLCs were resolved in opposite ways: with the preimage on one channel and by timeout on the other, so one party lost the HTLC amount (0.48 BTC in all). In 38 of them (0.44 BTC) the upstream HTLC timed out while the downstream one was paid, so the node between them paid out and was not paid; in 100% of those the preimage was already on-chain a median 58 blocks

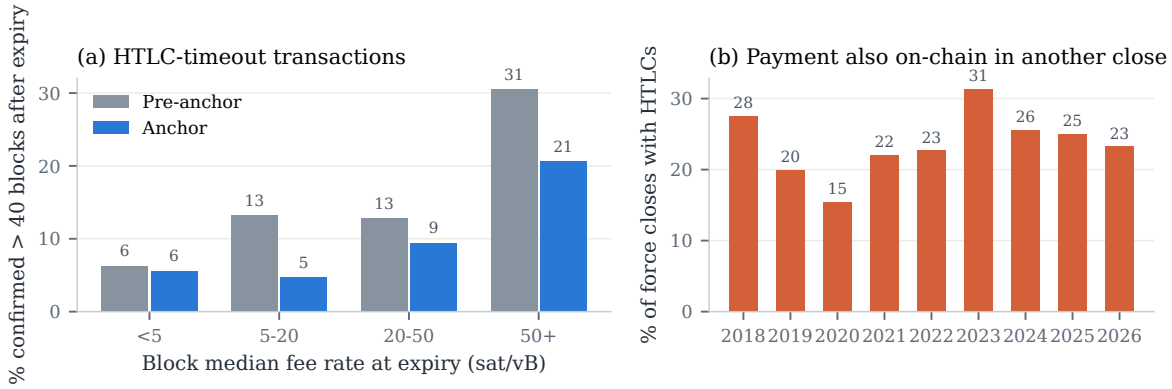


Fig. 4. HTLCs resolved on-chain. (a) Share of HTLC-timeout transactions confirmed more than 40 blocks after the HTLC's expiry, by the median fee rate of the block at expiry, for pre-anchor and anchor channels whose commitment had confirmed by the expiry. (b) Share of force closes with an HTLC resolved on-chain whose payment also appears in another force close, by year.

before the upstream timeout confirmed. The losses do not cluster in the 2023 fee spike, which points at nodes that were offline or failed to act rather than at fee competition. These counts leave out 7 pairs from 2026 (0.0016 BTC) whose channels used a 6-block `to_self_delay`: a replacement-cycling test, described next, not organic losses.

f) No loss has the order replacement cycling leaves: In a replacement-cycling attack the downstream peer keeps replacing the forwarder's HTLC-timeout transaction in the mempool, and lets its own preimage claim confirm only after the upstream HTLC has timed out [6]. A successful attack therefore puts the preimage on-chain after the upstream timeout. No forwarder loss from 2021 has that order. The staged pairs do, and they are such an attack, run as a test: one operator ran the upstream, forwarding and downstream nodes and cycled its own forwarder's HTLC-timeout transactions on mainnet. Shared funding and direct transfers link the three nodes' wallets, and the transactions that evicted the forwarder's timeouts carry a fingerprint we find nowhere else on the chain in that period. In each staged pair the preimage confirmed no earlier than the upstream timeout, a median of 4 blocks after. The test sees only payments whose two HTLCs were both resolved on-chain, so it bounds attacks that succeeded that way, not attempts.

D. The whole life of a channel

a) A channel's life costs a few thousand satoshis: Funding, closing and the spends a force close requires cost a public channel a median of 3,435 sat (Table III): 723 for its share of the funding transaction and 1,411 for the close. The median life cost 0.15% of the channel's capacity. The fee regime dominates. Channels closed in 2019–20 paid the most, a median 10,510 sat; those closed in 2023–24 paid 6,483, and those closed in 2025–26 986.

b) The open is a large part of the bill: Opens paid 40% of all lifecycle fees (28.2 of 70.9 BTC), and 63% for the channels closed in 2025–26, which closed cheaply but had often opened in dearer blocks (Fig. 5b).

c) Counting the open narrows the gap between endings: A life that ended cooperatively cost a median 1,970 sat, one that ended in a force close 6,645, a ratio of 3.4 (1.3–3.9 by year), against 2–6 for the close alone, because both kinds of channel paid the same to open. If change outputs also paid their share of the funding fee, by value or by size, the median life would cost 18–19% less; treating funding transactions that also pay others as payment batches moves it by 1%. The lifecycles are completed ones, so they favour short-lived channels (median life 75 days).

VII. What Anchor Outputs Are Used For

a) No P2WSH anchor is left unspent: All 387,502 P2WSH anchors created by force closes, public or not, at least four weeks before the cutoff had been spent by the cutoff (Table IV). Anchors that their owners leave behind do not stay in the UTXO set: third parties sweep them. That is the premise of an open LND proposal to stop sweeping them [16]. For P2WSH anchors the data confirm it; simple-taproot anchors, below, are the exception.

b) Owners stopped using their anchors: Owners spent 81% of the anchors of 2022's force closes and 5% of 2026's (Fig. 6a, Table IV). Most owner spends were cleanup after the commitment had confirmed, not fee bumps. Bumps peaked in 2023, when owners bumped 32% of anchor force closes without pending HTLCs and 78% of those with them. Even a same-block child is often not a bump: in 2026, 70% of them did not raise the package's fee rate, so they were cleanup that confirmed in the same block.

c) Sweepers wait for cheap blocks when fees are high: Anyone may sweep an anchor from the sixteenth block after the close, but when fees were high nobody did so at once. The median third-party sweep of a public channel's anchor came 1,708 blocks after a 2023 close, about 12 days, against exactly 16 from 2025 (Fig. 6b). For 2023 closes, 85% of sweeps confirmed in a block cheaper than the one sixteen blocks after the close (median fee rate 5 against 13 sat/vB). The anchors of 2021's closes waited for the cheap blocks of 2022.

TABLE III

WHAT A PUBLIC CHANNEL'S LIFE COSTS ON-CHAIN: FUNDING SHARE, CLOSE AND REQUIRED SPENDS, BY YEAR OF CLOSE (MEDIANS, SAT UNLESS STATED).

Closed	Channels	Force %	Open	Coop.	Force	All	Range	p90	% cap.	Opens %	BTC
2019	60,405	49.7	2,429	8,322	11,021	9,581	7,842–9,581	28,253	1.29	36.3	8.00
2020	46,598	38.2	2,362	9,061	19,174	12,750	10,708–12,750	38,299	1.51	36.6	8.55
2021	91,441	30.0	925	2,127	8,329	3,368	2,701–3,368	30,606	0.39	47.0	10.11
2022	150,542	38.5	361	1,162	4,249	2,153	1,790–2,153	7,859	0.10	44.9	6.04
2023	112,481	44.5	1,069	3,864	10,119	6,106	5,090–6,106	33,969	0.22	30.0	20.51
2024	79,763	36.2	2,290	5,544	10,710	7,032	5,437–7,032	31,803	0.18	44.3	14.07
2025	79,656	27.0	477	907	2,374	1,261	937–1,261	5,331	0.03	62.0	2.40
2026[a]	71,639	20.1	309	626	1,450	785	569–785	2,927	0.01	64.6	1.24
All	692,525	35.8	723	1,970	6,645	3,435	2,804–3,435	23,171	0.15	39.7	70.92

[a]Closes to 4 October 2026 (four weeks for outputs to be spent). Open: the channel's share of its funding transaction's fee, split equally over the channel outputs of the transaction; other outputs are the funder's change and pay nothing. Range: the median lifecycle if every output, change included, pays for its own size, and with the equal split. Coop., Force: lifecycles ending in a cooperative or a force close; force closes as in Table II. % cap.: median lifecycle fees over capacity. Opens %: opens' share of all lifecycle fees. BTC: all lifecycle fees. Closes from 2019: our list of public channels holds few of the channels that closed in 2018.

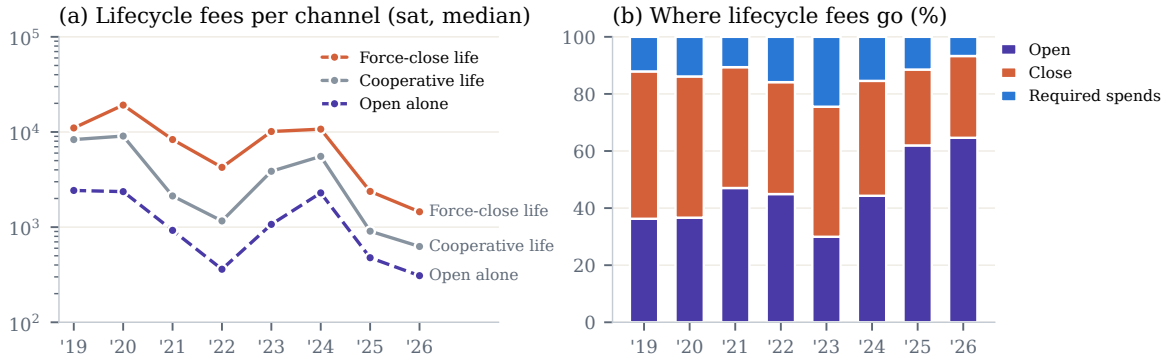


Fig. 5. A public channel's on-chain fees from funding to its last required spend. (a) Median by year of close, for lives ending in a force close or a cooperative close, and the channel's share of its funding transaction alone (log scale). (b) Where lifecycle fees go: the open, the close and the spends a force close requires.

TABLE IV

WHAT ANCHOR OUTPUTS ARE USED FOR. LEFT: ANCHORS BY THE YEAR OF THEIR FORCE CLOSE. RIGHT: THIRD-PARTY SWEEP TRANSACTIONS BY THE YEAR THEY CONFIRMED.

Year	Anchors	Owner %	Third %	Unspent %	Bumped %	Delay	Sweep txs	Per tx	sat/vB	Miners %
2021	6,674	93.7	6.3	0.0	8.3	15,054	–	–	–	–
2022	62,826	81.3	18.7	0.0	12.5	78	1,341	9.1	1.00	23.9
2023	70,131	35.2	64.8	0.0	43.8	1,708	1,117	37.3	3.44	65.3
2024	68,631	15.3	84.7	0.0	18.3	3,308	3,021	21.7	4.90	84.1
2025	102,663	11.9	88.1	0.0	9.0	16	9,474	9.5	2.50	82.6
2026[a]	76,577	4.8	95.2	0.0	13.6	16	10,390	7.4	1.66	87.0

[a]Closes to 4 October 2026. Anchors: P2WSH anchors of all force closes, public or not. Bumped: share of public anchor force closes with an owner child in the commitment's block. Delay: median blocks from the close to a third-party sweep. Sweep txs: transactions spending anchors with an empty signature (chain-wide). sat/vB: their median fee rate. Miners: share of the anchors' value paid as fees. Break-even fee rate: 6.38 sat/vB.

d) Miners take most of the value: Fees took 24% of the swept anchors' value in 2022, when sweeps paid a median 1.0 sat/vB, and 83–87% from 2024 (Fig. 6c). From 2022 to the cutoff, third-party sweeps collected anchors worth 0.95 BTC; 0.75 BTC of it went to miners and 0.20 BTC to the sweepers. A sweep pays only below 6.38 sat/vB, which is why sweepers wait out high fees.

e) One operator collects most third-party anchors: We find 332 sweeping operators from 2022 to the cutoff. Sweeps that give everything to miners, with only `OP_RETURN` outputs, took 10% of third-party anchors; of the rest, 15 operators took more than 1% each. The largest appeared on 27 July 2024 and took 58% of third-party anchors in 2026 and 41% of all since 2022, netting 0.070 BTC.

f) No third party has swept a simple-taproot anchor: Simple taproot channels' anchors mostly stay unspent. Of 23,864 created from 2023 to the cutoff, 1,706 were spent; of the 20,177 created in 2026, 71. Every one of those spends is a key-path spend by the anchor's owner, and no input anywhere on the chain spends the anchor's 16-block script path (0 inputs). That path needs the anchor's internal key, which the commitment does not reveal; the specification has

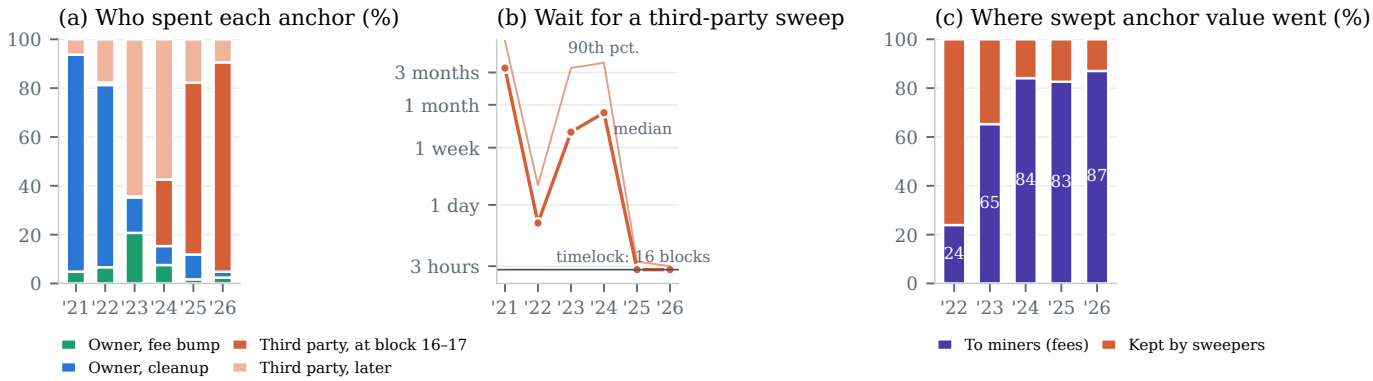


Fig. 6. The anchor money trail. (a) Who spent each P2WSH anchor, by year of the force close: its owner in the commitment's block (fee bump) or later (cleanup), or a third party at blocks 16–17 or later. (b) Wait from the close to a third-party sweep, public channels: median and 90th percentile (log scale). (c) Where swept anchor value went, by year of the sweep: fees to miners, or kept by sweepers.

it revealed when the matching `to_local` or `to_remote` output is swept [32]. The missing key does not explain the unspent anchors: for 14,828 of the 22,158 unspent anchors the key has been on-chain for a median of 101 days, and the sweepers that clear P2WSH anchors have not touched them. Why they do not is a question for the implementations' developers.

g) The unswept taproot anchors are worth little to a sweeper: The anchors whose key is on-chain hold 0.049 BTC in all, less than the largest P2WSH sweeper has netted (0.070 BTC). Owners' own spends stopped as well, even though adding the anchor to a sweep still paid: in the first half of 2025, 96% of owners' `to_remote` sweeps also spent the matching anchor, against 0.2% from July 2026 to the cutoff. The 2026 sweeps paid a median 0.9 sat/vB, against 2.0 sat/vB in early 2025, so including the anchor was worth more to the owner in 2026, not less.

VIII. Fee Regimes

A. Fee spikes do not bring a wave of force closes

a) No wave: Fee spikes did not bring a wave of force closes. Pooled over the 9 spikes, organic public force closes in the spike week and the week after were 0.97 times their level in the four weeks before (Table V). After random ordinary weeks of the same years the ratio is typically 1.02, so the spikes sit at percentile 23 of those draws, below a typical week rather than above it. A wave 15% above the typical ratio would have been detected. Pre-anchor channels, whose fee is fixed at signing and which an `update_fee` disagreement would close in a spike, show the same (0.98), as do anchor channels (0.94). Within years, the elasticity of the weekly force-close rate to the median fee rate of the same week is 0.00, at percentile 55 of a placebo that moves the fee series in time.

b) The spikes themselves: The spikes went both ways. 4 of the 9 came with more force closes, the most in May 2023 (29% more, mostly in anchor channels), when the median fee rate went to 84 sat/vB from 13. Next came March 2019, with 26% more, all in pre-anchor channels and in a month when our channel list may still have been filling in. The other 5 came with fewer force closes, as many as 50% fewer in April 2024 (Fig. 7).

TABLE V
Fee spikes against placebos. Top: organic public force closes in a window after the spike week, divided by the four weeks before, pooled over the spikes, against random ordinary weeks of the same years. Bottom: elasticity of the weekly organic force-close rate to the median block fee rate, within years, by the timing of the fee rate, against placebos that move the fee series in time.

Weeks after spike	Spikes	Ordinary	90% range	Pctl.
All public, 0 to +1	0.97	1.02	0.90–1.17	23
Pre-anchor, 0 to +1	0.98	1.00	0.89–1.19	42
Anchor, 0 to +1	0.94	1.04	0.86–1.26	17
All public, +2 to +3[a]	1.10	1.01	0.90–1.14	88
Fee timing	b	**Null 95% (C)**	**Pctl. (C)**	**Pctl. (B)**
1 week later (lead)	0.02	−0.08 to 0.07	69	72
Same week	0.00	−0.08 to 0.07	55	55
1 week earlier	0.01	−0.08 to 0.08	61	63
2 weeks earlier	0.02	−0.08 to 0.07	67	76
3 weeks earlier	0.01	−0.08 to 0.07	65	69
4 weeks earlier	0.00	−0.07 to 0.07	46	51
Largest, 1–4 earlier	0.02	−0.06 to 0.08	58	64

Ordinary: median ratio over 10,000 draws of one ordinary week per spike from the same year; 90% range of those draws. Pctl.: percentile of the real value among the draws. (C): circular shift of the fee series, all 382 shifts; (B): 8-week blocks shuffled within each year, 5,000 draws. [a]Window chosen after seeing the lag profile.

c) No later rise either: Closes broadcast during a spike at too low a fee would confirm only in the weeks after it, and the chain shows confirmation, not broadcast. Those weeks show no rise beyond placebo noise. With the fee rate of one to four weeks earlier, the elasticity is at most 0.02; tested over the four lags together, $p = 0.420$ against the circular shift and $p = 0.359$ against the block shuffle. In the event windows, weeks +2 to +3 reach 1.10 times the weeks before, at percentile 88 of ordinary weeks.

d) Unlisted closes move the other way: Unlisted force closes fell when fees rose (elasticity −0.22). Mass closes cannot be removed from them, because their nodes are unknown, so we do not interpret this.

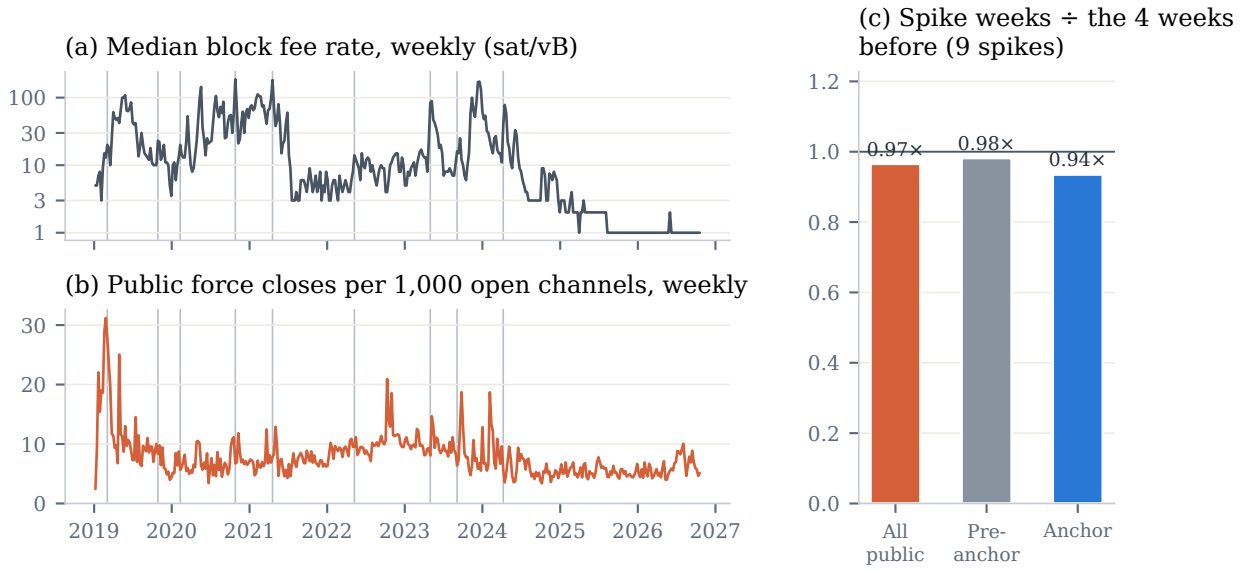


Fig. 7. Fee spikes and force closes. (a) Weekly median block fee rate (log scale); vertical lines mark the spike weeks. (b) Weekly public force closes per 1,000 open channels, mass-close events excluded. (c) Force closes in the spike week and the week after, divided by those in the four weeks before, pooled over the 9 spikes.

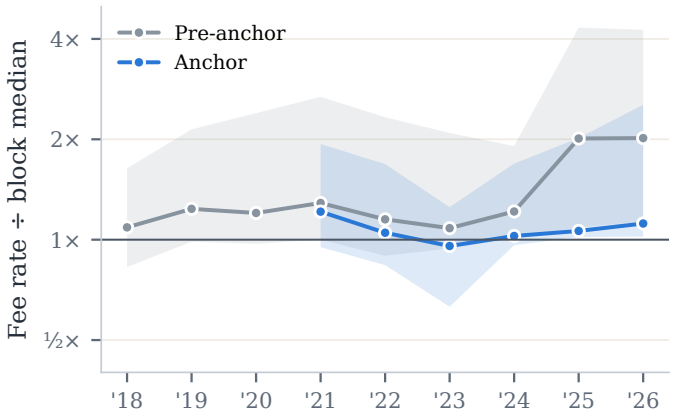


Fig. 8. A commitment's fee rate divided by the median fee rate of the block it confirmed in, by year: median and interquartile range, pre-anchor and anchor commitments.

TABLE VI
LIGHTNING'S CLOSING BILL: FEES PAID BY CHANNEL CLOSES AND THE TRANSACTIONS THEY REQUIRE, PUBLIC OR NOT, AGAINST ALL BITCOIN TRANSACTION FEES AND BLOCK SPACE.

Year	Coop. BTC	Force BTC	Total BTC	% of fees	% of block space
2018[b]	–	1.39	–	–	–
2019	1.74	5.44	7.18	0.04	0.04
2020	2.12	5.14	7.25	0.03	0.03
2021	2.19	4.91	7.10	0.03	0.05
2022	0.82	3.29	4.11	0.08	0.12
2023	2.40	14.43	16.83	0.07	0.10
2024	1.88	9.55	11.43	0.08	0.08
2025	0.28	2.09	2.37	0.14	0.10
2026[a]	0.23	1.29	1.52	0.18	0.11

[a]January–October. Force: commitments, required spends (split per input) and third-party anchor sweeps; cooperative closes of channels outside our public list cannot be seen and are not included, nor are channel opens. [b]From 12 January; cooperative closes of 2018 are mostly outside our list, so only force closes are shown.

B. Fixed commitment fees overpaid in calm markets

a) Pre-anchor commitments paid twice the market when fees were low: Pre-anchor commitments paid a median 1.1–1.3 times the median fee rate of their own block from 2018 to 2024, and 2.0 times in 2025–26 (Fig. 8). In 2026, 52% paid more than double, against 32% in 2021. Over the period, pre-anchor commitments paid 10.9 BTC more than the tenth percentile of their blocks. A commitment that paid too little waits for a cheaper block instead, so this measure shows overpayment; underpayment shows up as delay.

b) Anchor commitments paid about the market: Anchor commitments paid a median 0.96–1.21 times their block's median. In 2023, 27% confirmed below their block's tenth percentile, that is, with a child's help.

C. Lightning's closing bill is small

a) A fraction of a percent of Bitcoin's fees: All closes and the spends they require, public or not, with third-party anchor sweeps, paid 16.8 BTC in 2023, the most in any year, and 1.5 BTC in the first ten months of 2026 (Table VI). In every year that was 0.03–0.18% of all transaction fees and at most 0.12% of block space. The share of fees rose in the low-fee years because everything else became cheaper faster.

IX. Zero-Fee Commitments

a) The first closes came before the specification: The first zero-fee commitment we find closed a private channel on 29 October 2025, and 6 had closed before the specification merged the format in May 2026. The first public one closed on 28 September 2026. By the cutoff there were 31: 18 private and 13 public, involving 10 distinct nodes, one of them party to

TABLE VII
BLOCK SPACE OF THE ANCHOR MECHANISM PER FORCE CLOSE, OBSERVED (TWO P2WSH ANCHORS AND THEIR SPENDS) AND UNDER ZERO-FEE COMMITMENTS (ONE PAY-TO-ANCHOR OUTPUT SPENT BY A CHILD).

Year	Anchor closes	Bumped %	Anchors vB	P2A vB	Anchor BTC
2021	3,839	8.4	194.1	155.9	0.022
2022	34,297	12.2	200.4	151.8	0.207
2023	39,280	36.6	180.5	124.6	0.231
2024	41,035	12.6	163.1	151.3	0.226
2025	59,927	2.9	166.0	162.0	0.339
2026[a]	45,209	4.1	162.0	160.7	0.253

[a]Closes to 4 October 2026. Zero-fee closes that were not already bumped need a child of their own (wallet input and change output, 111.0 vB). Anchor BTC: value locked in 330-sat anchors, which pay-to-anchor outputs (0 sat) do not create.

every public close. Every commitment paid no fee, and every pay-to-anchor output was spent by a child in the same block, at a median package fee rate of 1.6 sat/vB. The channels were small (median capacity 49,155 sat).

b) The new format saves little block space: Across the 223,587 anchor force closes we observe, the anchor mechanism took 173 vB per close: two anchor outputs and the transactions that spent them. Zero-fee commitments would have taken 152 vB: one pay-to-anchor output, its spend, and a child for the 88% of closes that were not already bumped (Table VII). That is 12% less. The gains lie elsewhere: the 1.28 BTC held in 330-sat anchor outputs would not exist, commitment fees no longer have to be guessed at signing, and package relay replaces the carve-out that Bitcoin Core has removed [7], [10].

X. DISCUSSION

a) Fee spikes do not break channels: A common view holds that fee spikes break Lightning channels: commitment fees signed in calm markets fall behind, peers disagree over `update_fee`, and channels are forced closed in a wave. In confirmed force closes we see no such wave, for pre-anchor channels as little as for anchor channels. 4 of the 9 spikes did bring more force closes, the most in May 2023, but pooled they sit below ordinary weeks of the same years. Spikes still matter, but through what happens after a close: HTLC transactions confirm later, owners bump commitments, and sweepers wait. The caveat is that the chain records confirmation, not broadcast. A close broadcast in a spike at too low a fee may wait for a cheaper block, which would spread a wave of broadcasts over the following weeks; those weeks show no rise beyond placebo noise either.

b) Where the anchor design worked, and where it did not: Anchor outputs were meant to let the broadcaster pay the fee the market asks at broadcast time. For commitments that is rarely needed: outside 2023, owners seldom bumped, and in 2026 most same-block owner children did not raise the fee rate at all. Where the design shows its value is in HTLC transactions. In fee spikes, anchor channels' HTLC timeouts were late far less often than pre-anchor ones, and late HTLCs are what costs a forwarding node money. Meanwhile the anchors themselves became third-party dust: owners leave them, sweepers collect them at the first block or wait for cheap blocks, and miners take most of their value. This is the behaviour LND's open issue on dangling anchors assumes [16], measured, but only for P2WSH anchors. No third party sweeps simple-taproot anchors, and their owners rarely do: of the 20,177 created in 2026, 71 were spent. Each taproot force close therefore leaves outputs in the UTXO set, which a zero-value pay-to-anchor output, spent together with its commitment, does not (§IX).

c) The migration: Zero-fee commitments change little in block space. Their gains are a simpler fee story, no 330-sat outputs to sweep, and relay rules that no longer depend on the removed carve-out. Their test is the one on which anchors improved on the old format: whether HTLC transactions confirm before their deadlines when fees spike. Our measurements are the baseline for it: at 50 sat/vB or more, 20.6% of anchor channels' HTLC timeouts confirmed more than 40 blocks late. How fast these gains arrive depends on operators, not on the specification: a channel keeps its format until it closes. Outside one cluster of short-lived channels, pre-anchor commitments were 8% of 2026's public force closes, in channels a median of 3.9 years old, years after anchors became available. Our second paper found that most Lightning operators upgrade within months of a release while a tail stays far behind [18], so the old formats are likely to stay on-chain for years. The same choice, between fixing a fee when a transaction is signed and paying it when it is broadcast, faces every protocol built on pre-signed transactions and timelocks, and there too what matters is whether the time-critical transactions confirm on time [5].

d) What a force close costs: In fees, a force close is cheap and a channel's life costs a fraction of a percent of its capacity, and the open is a large part of that. The cost of a force close is a ratio, a multiple of a cooperative close in fees and block space, and a wait, days before the closer's own funds return; the wait outweighs the fees only in low-fee years. Its exposure, the HTLCs that must confirm before their deadlines while fees are high, has rarely become a loss: since 2021, forwarding nodes lost 38 payments that way, each with the preimage already on-chain.

e) What the chain can catch: Two 2026 events show what transactions alone can reveal. A week of revoked commitments matched a splicing bug that Core Lightning had fixed days before, and the only split resolutions in the order replacement cycling leaves were a test of that attack on mainnet. We found both without node logs, and the same tests can be run on new blocks as they arrive.

XI. LIMITATIONS

Broadcaster. The chain does not show which party broadcast a force close, so costs are per channel, both parties together. *Invisible bumps.* Fee bumps that never confirmed, and replaced transactions, leave no trace on the chain; we have no mempool record. *Private channels.* Private channels appear only when they force close, so their number and their force-close rate are

unknown, and their cooperative closes are not in the closing bill. Before 2024, "unlisted" also includes public channels missing from our list. *Open channels.* Channels whose nodes have left count as open until closed on-chain, which lowers the per-channel rate as they accumulate; cohort incidence is not affected. *Penalties.* Only punished breaches are visible. *Scope of the cost account.* Pre-anchor `to_remote` spends are excluded by design, and third-party anchor sweeps are counted as value lost, not as fees the channel pays. The break-even rate counts the broadcaster's locked balance only, not pending HTLC amounts. Simple taproot channels' spends are not classified, so their delays, HTLCs and bumps are missing from §VI; they were 23% of 2026's unlisted force closes. *Fee regimes.* Every result depends on the fee regime, which is why the window covers all of Lightning's history. Anchor channels appear in our data from 2021, so anchor results start then. *Lifecycle.* Opens are costed under an attribution rule; dual-funded opens are not identified (which does not change a per-channel cost), and only completed lives enter, so short-lived channels weigh more. *Client labels* come from our first paper, inherit its accuracy and start in May 2023. *Association.* The fee-spike test is a within-year association; it rules out a large same-week effect, not every causal path.

XII. Ethics

All inputs are public blocks and public gossip. We report aggregates only: no list, ranking or label of nodes, sweeping operators, addresses or transactions, in the paper or in the released data. In particular we do not identify the dominant anchor sweeper, the nodes of the first zero-fee commitment closes, the operator of the replacement-cycling test or the channels of the breach wave. The paper reports no new vulnerability: replacement cycling was disclosed in 2023 [6], and the splicing bug was fixed before the wave [31].

XIII. Conclusion

Fee spikes do not bring waves of Lightning force closes, for channels whose fee is fixed at signing or for those that can bump it. What a force close costs is a multiple of a cooperative close in fees and days of locked funds, and the wait outweighs the fees only in low-fee years. When fees are high, pending payments are exposed, yet forwarders lost few of them, each with the preimage already on-chain. The anchor design reduced that exposure for HTLCs, while its P2WSH anchors became dust that third parties sweep and miners collect, and its taproot anchors stay in the UTXO set. Zero-fee commitments will save little block space; the measure of the migration is whether payments still resolve before their deadlines when the next spike comes.

Appendix A
Force Share by Client Pair

Table VIII gives the force share of public closes since 8 May 2023 by the implementations at the two ends of the channel, from the labels of our first paper [17]. The chain does not show which side broadcast a force close, so a close is attributed to the pair, not to one implementation. Among pairs with both ends labelled and more than 1,000 closes, the force share ranges from 28% (LND–LND, 169,435 closes) to 40% (CLN–CLN); CLN–LND (35%), LDK–LND (37%) and Eclair–LND (29%) lie between. Pairs with an unlabelled end run higher, for example 44% for LND with an unlabelled peer. The one outlier, LDK–LDK at 1.5% of 13,415 closes, is one node closing channels cooperatively at scale, not a property of the implementation: a single node is party to 99.5% of those closes. For the other pairs we have not separated operators from implementations, so the table describes pairs and does not rank implementations.

TABLE VIII
Force share of public closes by the client pair at the two ends (closes from 8 May 2023).

Client pair	Closes	Force	Force %
LND–LND	169,435	47,905	28.3
CLN–LND	20,611	7,149	34.7
LDK–LND	17,749	6,603	37.2
LDK–LDK	13,415	204	1.5
Eclair–LND	11,474	3,335	29.1
CLN–CLN	1,832	740	40.4
CLN–Eclair	927	291	31.4

Clients from paper 01's labels on the day of the close (high or medium confidence). The chain does not show which side broadcast a force close, so a close is attributed to the pair.

Data and Code Availability

Code, the aggregate datasets behind every figure and table, and a script that recomputes every number quoted in the text are available at https://github.com/bitcoindatalabs/ln-force-close-costs. All stages truncate their inputs at the data cutoff, block 974,041 (31 October 2026). The released aggregates name no node, address or transaction. The underlying blocks are public; our channel list and the per-transaction tables are available from the author on request.

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